DEPARTMENT OF COMPUTER ENGINEERING
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ILE-IFE, NIGERIA.

CPE 204: Introduction to Digital Computer Systems II

Tutorial Questions for 2024/2025 Session (20 Marks)

1. (a) Use the universal logic gate (NAND) to implement the following logic gates.
   (i) AND gate
   (ii) OR gate
   (iii) NOT gate
   (iv) NOR gate
   (v) XOR gate
   (vi) XNOR gate

   (b) Use the universal gate (NOR) to implement the following logic gates.
   (i) AND gate
   (ii) OR gate
   (iii) NOT gate
   (iv) NAND gate
   (v) XOR gate
   (vi) XNOR gate

2. (a) Draw a T flip-flop and explain its basic operation.
   (b) Draw the D flip-flop and explain its basic operation.

3. (a). Design a 3-bit synchronous up counter.
   (b). Design a 4-bit asynchronous down counter.

4. (a). Design a Mod 13 ripple counter using J-K flip-flop.
   (b). Differentiate between Programmable Array Logic and Programmable Logic Array.

5. Design a logic circuit that controls an elevator door in a three-story building. *M* is a logic signal that indicates when the elevator is moving (*M* = 1) or stopped (*M* = 0). *F1*, *F2*, and *F3* are floor indicator signals that are normally LOW, and they go HIGH only when the elevator is positioned at the level of that particular floor. For example, when the elevator is lined up level with the second floor, *F2* = 1 and *F1* = *F3* = 0. The circuit output is the *OPEN* signal, which is normally LOW and will go HIGH when the elevator door is to be opened.

6. A manufacturing plant needs to have a horn sound to signal quitting time. The horn should be activated when either of the following conditions is met:
   i. It's after 5 o'clock and all machines are shut down.
   ii. It's Friday, the production run for the day is complete, and all machines are shut down.

Design a logic circuit that will control the horn. (*Hint:* Use four logic input variables to represent the various conditions; for example, input *A* will be HIGH only when the time of day is 5 o'clock or later.)

7. A four-bit binary number is represented as $A_3\ A_2\ A_1\ A_0$, where A_3, A_2, A_1, and A_0 represent the individual bits and A_0 is equal to the LSB. Design a logic circuit that will produce a HIGH output whenever the binary number is greater than 0010 and less than 1000.

8. An automobile alarm circuit is used to detect certain undesirable conditions. The three switches are used to indicate the status of the door by the driver's seat, the ignition, and the headlights, respectively. Design the logic circuit with these three switches as inputs so that the alarm will be activated whenever either of the following conditions exists:

i. The headlights are on while the ignition is off.
ii. The door is open while the ignition is on.

9. A *BCD counter* produces a four-bit output representing the BCD code for the number of pulses that have been applied to the counter input. For example, after four pulses have occurred, the counter outputs are $DCBA = 0100_2 = 4_{10}$. The counter resets to 0000 on the tenth pulse and starts counting over again. In other words, the *DCBA* outputs will never represent a number greater than $1001_2 = 9_{10}$.
   i. Design the logic circuit that produces a HIGH output whenever the count is 2, 3, or 9. Use K mapping and take advantage of the don't-care conditions.
   ii. Repeat for $x = 1$ when $DCBA = 3,4,5,8$.

10. Four switches are part of the control circuitry in a copy machine. The switches are at various points along the path of the copy paper as the paper passes through the machine. Each switch is normally open, and as the paper passes over a switch, the switch closes. Switches SW1 and SW4 cannot be closed at the same time. Design the logic circuit to produce a HIGH output whenever *two or more* switches are closed at the same time. Use K mapping and take advantage of the don't-care conditions.

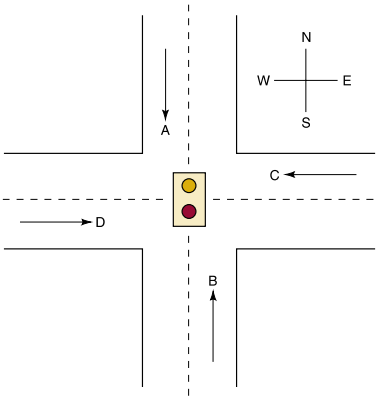
11. A BCD code is being transmitted to a remote receiver. The bits are A_3, A_2, A_1, and A_0, with A_3 as the MSB. The receiver circuitry includes a *BCD error detector* circuit that examines the received code to see if it is a legal BCD code (i.e., ≤ 1001). Design this circuit to produce a HIGH for any error condition.

12. Four large tanks at a chemical plant contain different liquids being heated. Liquid-level sensors are being used to detect whenever the level in tank *A* or tank *B* rises above a predetermined level. Temperature sensors in tanks *C* and *D* detect when the temperature in either of these tanks drops below a prescribed temperature limit. Assume that the liquid-level sensor outputs *A* and *B* are LOW when the level is satisfactory and HIGH when the level is too high. Also, the temperature-sensor outputs *C* and *D* are LOW when the temperature is satisfactory and HIGH when the temperature is too low. Design a logic circuit that will detect whenever the level in tank *A* or tank *B* is too high at the same time that the temperature in either tank *C* or tank *D* is too low.

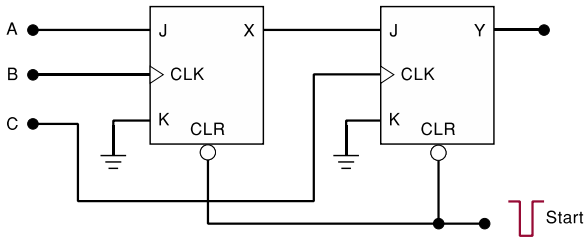
13. The figure below shows the intersection of a main highway with a secondary access road. Vehicle-detection sensors are placed along lanes *C* and *D* (main road) and lanes *A* and *B* (access road). These sensor outputs are LOW (0) when no vehicle is present and HIGH (1) when a vehicle is present. The intersection traffic light is to be controlled according to the following logic:

   i. The east-west (E-W) traffic light will be green whenever *both* lanes *C* and *D* are occupied.
   ii. The E-W light will be green whenever *either C* or *D* is occupied but lanes *A* and *B* are not *both* occupied.
   iii. The north-south (N-S) light will be green whenever *both* lanes *A* and *B* are occupied but *C* and *D* are not *both* occupied.
   iv. The N-S light will also be green when *either A* or *B* is occupied while *C* and *D* are *both* vacant.
   v. The E-W light will be green when *no* vehicles are present.

Using the sensor outputs *A*, *B*, *C,* and *D* as inputs, design a logic circuit to control the traffic light. There should be two outputs, N-S and E-W, that go HIGH when the corresponding light is to be *green.* Simplify the circuit as much as possible and show *all* steps.



14. In the circuit below, inputs *A*, *B*, and *C* are all initially LOW. Output *Y* is supposed to go HIGH only when *A*, *B*, and *C* go HIGH in a certain sequence.
    (a) Determine the sequence that will make *Y* go HIGH.
    (b) Explain why the START pulse is needed.
    (c) Modify this circuit to use D FFs.



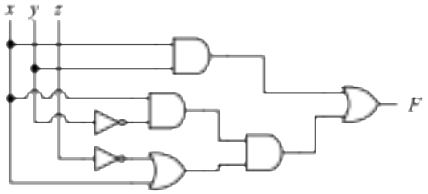
15. (a) Draw a circuit diagram for the synchronous parallel transfer of data from one three-bit register to another using J-K flip-flops.
    (b) Repeat for asynchronous parallel transfer.

16. a. How many FFs are required to build a binary counter that counts from 0 to 1023?
(b) Determine the frequency at the output of the last FF of this counter for an input clock frequency of 2 MHz.
(c) What is the counter's MOD number?
(d) If the counter is initially at zero, what count will it hold after 2060 pulses?

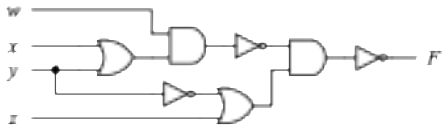
17. (a) Design a Half Adder.
    (b) Design a Full Adder.

18. Derive the truth table for the following circuits:

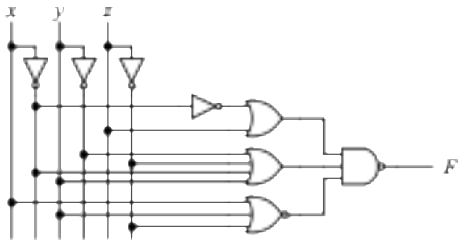
a)



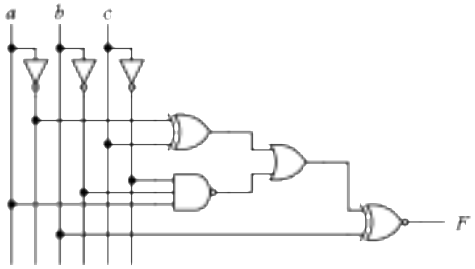
b)



c)



d)



19. Draw the circuit diagram that implements the following expressions:
    a) $F(x, y, z) = \Sigma(0, 1, 6)$
    b) $F(w, x, y, z) = \Sigma(0, 1, 6)$
    c) $F(w, x, y, z) = \Sigma(2, 6, 10, 11, 14, 15)$
    d) $F(x, y, z) = \Pi(0, 1, 6)$
    e) $F(w, x, y, z) = \Pi(0, 1, 6)$
    f) $F(w, x, y, z) = \Pi(2, 6, 10, 11, 14, 15)$

20. Design a BCD to 7-segment display code converter.

21. A computer crashes due to incorrect instruction sequencing. Apply your knowledge of the instruction cycle to identify possible causes.

22. Explain the functions of the following CPU registers:
    (a) Program Counter (PC)
    (b) Memory Address Register (MAR)
    (c) Memory Data Register (MDR)
    (d) Accumulator (AC)

23. With the aid of block diagram, explain in details Von Neumann Architecture.

24. When a computer system stores both data and instructions in the same memory. Explain how the stored program concept enables this system to execute different programs without hardware modification.

25. Explain the roles of MAR, MDR, ALU, and Accumulator in instruction execution of CPU carrying out simple arithmetic

Due Date: July 20, 2026
Time: 12:00pm